Uniquea: A Request for Audit

To: UC San Diego, Department of Physics
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Date: October 18, 2026
Length: two pages. A full archive (about 2.5 MB) and an audited core (about 25 pages) are available on request.

What we are asking

We are not asking you to endorse anything. We are asking for an audit: check the mathematics below, and tell us which of the proposed tests fail. A clear "this is wrong, and here is why" is a useful answer.

Who did what

The ideas, the observations and the framework's faith statements are ours. Most of the formal mathematics was drafted by AI assistants (Grok, Gemini, Meta, Manus and Claude) at our direction, then checked and re-run by Claude using CODATA constants. We did not design the formal "Planck 1500 Ladder" model ourselves. Each claim below carries a status label:

- **Established:** accepted physics.
- **Derived:** follows from stated assumptions by checkable steps.
- **Postulated:** our proposal, not derived from anything deeper.
- **Candidate:** a proposal that data could support or refute.
- **Open:** asked, not answered.

The framework in five lines (Postulated)

1. Space is a plenum: a dynamic graviton field fills it, and changes in the field travel at c.
2. The universe is cyclic. It contracts to a cold, dense state and rebounds without a singularity (the "Cold Crunch").
3. A small number of primordial black holes survive the rebound already massive.
4. These black holes add new field quanta, and so contribute to expansion.
5. A family of force laws (the "Nash ladder," a project name unrelated to John Nash's work) may replace dark matter at low acceleration.

Three results that survive checking

1. A self-sustaining cycle (Derived, re-run). A four-variable system built on a Stuart-Landau oscillator has a stable limit cycle. Its Floquet multipliers are 1, 0.0432, 0.00187 and 3.49×10^{-6}, confirmed from three starting points. A minimal closure, $H = \omega_o q$, gives a scale factor that never reaches zero. *Open:* nothing yet ties the model's variables to a measured quantity, and the source of the oscillator's energy is unstated.

2. The modified couplings reduce to Einstein (Derived). Our variants, such as $8\pi^2G/c^8$ and $8\pi^3G/c^{16}$, each need a dimensioned bridge constant. Choosing the bridge to match tested gravity returns $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ exactly. So the only form that could carry new physics is

$$G_{\mu\nu} + \Lambda_{eff}\, g_{\mu\nu} = (8\pi G/c^4) \cdot N \cdot T_{\mu\nu}\,, \quad N = 1 + (\text{small, dimensionless, varying})$$

A constant N would be absorbed into the measured G and could not be seen.

3. Rung zero is the deep-MOND law (Candidate). Written dimensionlessly, the ladder is

$$(F/F^*)^{(2^n)} = (m/m^*) \cdot (a/a^*)^{(2^{n+1})}$$

with Newton's F = ma kept separately as the seed. At n = 0 this gives $F = m \cdot a^2/a_o$, Milgrom's 1983 deep-MOND form. For reference, $cH_o/2\pi \approx 1.04 \times 10^{-10}$ m/s², close to the fitted $a_o \approx 1.2 \times 10^{-10}$ m/s². This is a lead, not a derivation.

Tests, stated so they can fail

#	Test	What would refute it
1	One a_0 fits galaxy rotation curves (rung zero)	Rotation-curve data (e.g. SPARC) need different a_0 values
2	A varying N (result 2 above)	Solar-system, pulsar or lensing data show no variation where the model predicts it
3	Exactly 36 primordial black holes	Far more than 36 objects of the QSO1 type. Little red dots at ~4×10^{-4} per Mpc^3 already strain this
4	Black-hole feeding by a "tether"	A rate fixed in advance fails on the next high-redshift black hole
5	Field creation drives expansion	The creation rate $\xi = 1$ only copies Λ. A $\xi \neq 1$ must predict a w(z) curve that DESI or Euclid can reject

Problems we have not solved

We list these so you don't have to find them:

- **Cosmic microwave background:** no derivation of the 2.725 K blackbody or its acoustic peaks from a *cold* rebound. A rebound hot enough for nucleosynthesis may be required.
- **Primordial helium (~25%):** no calculation.
- **Entropy across cycles (Tolman's problem):** no mechanism for removing entropy at the crunch.
- **Lensing and the Bullet Cluster:** not yet addressed by the ladder.
- **Isotropy:** the CMB is uniform to one part in 10^5, which constrains any edge or centre.
- **Ladder domain:** above rung zero the force barely depends on mass (2.2% change for doubled mass at n = 5). The ladder needs a stated domain.

Ideas we have already retired

Instantaneous gravity; a stationary lattice; a shared-seed explanation of entanglement (it obeys the Bell bound $|S| \leq 2$, and quantum mechanics reaches $2\sqrt{2}$); outflow from black holes; a "tired light" reading of redshift.

What we are not claiming

The framework sits inside a faith: we believe God created the universe for mankind to marvel at and measure. We keep that on its own shelf, labelled Faith, and we do not offer it as evidence. The physics above should stand or fall on its own.

Questions for the reviewer

1. Is result 3 worth fitting to rotation-curve data, or does a known problem already rule out rung zero?
2. Is there any form of varying N not already excluded by solar-system and pulsar tests?
3. Can a cyclic model with a cold rebound produce the CMB spectrum at all, or is a hot phase unavoidable?
4. Which of the five tests would you run first?

Thank you for your time.

Stephen M. Griffin